



Integrated Development Environments

Introduction to Programming

Now that we have view the introduction to programming video ([IntroProgramming_hb.mp4](#)),

Let's look at the big picture

Software Engineering

IDE

Integrated Development Environment (IDE)

All the software tools needed to create a program

IDE

Hides the details of the build process which is often complicated

Gives the appearance of a single tool

Graphical with many button shortcuts

Color coding of text known as syntax highlighting

Auto completion : guess what you intend

IDE Components

Code editor with syntax highlighting and auto completion

Compiler/Linker to build your code

Debugger/Interpreter to test your code

Alternative Method

Command line tools

Explicitly define the build process

Usually control the build process using “make”

User writes code in text editor such as vi, or emacs

A “Makefile” calls the compiler and linker to build the program

Alternative Method

IDEs hide the details of the alternative method

Makefile are problematic due to lack of agreement on how to check for individual computer capabilities

A huge mistake!

Lead to the creation of many different build systems

IDEs in this class

Visual Studio Code aka VSCode

Arduino

PlatformIO

Rust

CLion

Unix

Microsoft users will download WSL
or Windows Subsystem for Linux

Mac users need to install Xcode

Download NoMachine